1. Method study and
2. Work measurement.

Work study= Method study + Work measurement

Method Study

Method study is a systematic recording and critical analysis of the method of doing a work. It also proposes a new method of doing the same work in a easy and effective manner to reduce costs.

Work Measurement

Work measurement is a technique to find out the time taken for a qualified worker to finish a specified work at a particular level of performance.

11.6 PRODUCTION AND PRODUCTIVITY

Production can be defined as a process of manufacturing the required end product from the available raw materials.

Raw material → production → products

Productivity can be defined as a ratio between the output in a quartum of wealth and the input resources of production.

$$\text{Productivity} = \frac{\text{Production Output}}{\text{Input of Resources}}$$

11.7 PRODUCTIVITY BASED ON DIFFERENT RESOURCES- PRODUCTIVITY OF LAND

Let us assume an agriculturist spends ₹5000/- to plant casuarina saplings in his land of 1 acre and earns yields worth ₹10,000/-.

$$\text{Productivity} = \frac{\text{Production Output}}{\text{Input of Resources}}$$

$$= \frac{10000}{5000} \times 100$$

$$\text{Productivity} = 200\%$$

we consider another case where the same agriculturist spend. ₹6000 for better saplings and better methods of cultivator in the same area and earns yields worth ₹15,000/-.

$$\text{Productivity} = \frac{\text{Production Output}}{\text{Input of Resources}}$$

$$= \frac{15000}{6000} \times 100$$

$$= 250\%$$

In this case, the productivity is increased by 50%.

PRODUCTIVITY OF MEN

We consider a case of machinist who works on a milling machine and makes 40 gears a day. By improved methods of work, the same machinist is able to machine 50 gears in a day. The increase in productivity can be calculated as,

$$\text{Increase in Productivity} = \frac{\text{Change in Production}}{\text{Original Production (Input)}}$$

$$= \frac{50 - 40}{40} \times 100$$

$$= 25\%$$

Increase in Production = 25%

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